

→ Express

Library v/s Framework

Library

A library is a collection of pre-written code that can be used to perform specific tasks.

eg - axios

Framework

A framework is a set of pre-written code that provides a structure for developing software applications.

eg - express

Express

A Node.js web application framework that helps us to make web applications. It is used for **server** side programming.

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1. listen for incoming requests
2. parse
3. match response with routes
4. response



Getting Started with Express

```
const express = require("express");
const app = express();

let port = 8080;

app.listen(port, () => {
  console.log(`app listening on port ${port}`);
});
```

***Ports** are the logical endpoints of a network connection that is used to exchange information between a web server and a web client.

```
shradhakhapra@Shradhas-Air ExpressDir % npm init
This utility will walk you through creating a package.js
on file.
It only covers the most common items, and tries to guess
sensible defaults.
```

```
See 'npm help init' for definitive documentation on these
fields
and exactly what they do.
```

```
Use 'npm install <pkg>' afterwards to install a package
and
save it as a dependency in the package.json file.
```

```
Press ^C at any time to quit.
```

```
package name: (expressdir)
```

```
version: (1.0.0)
```

```
description:
```

```
entry point: (index.js)
```

```
test command:
```

```
git repository:
```

```
keywords:
```

```
author: Shradha
```

```
license: (ISC)
```

```
About to write to /Users/shradhakhapra/WebDevelopment/Backend/ExpressDir/package.json:
```

```
{
  "name": "expressdir",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "author": "Shradha",
  "license": "ISC"
}
```

```
Is this OK? (yes)
```

```
shradhakhapra@Shradhas-Air ExpressDir %
```

```
shradhakhapra@Shradhas-Air ExpressDir % npm install express
```

```
added 58 packages, and audited 59 packages in 1s
```

```
8 packages are looking for funding
run 'npm fund' for details
```

```
found 0 vulnerabilities
```

```
shradhakhapra@Shradhas-Air ExpressDir % touch index.js
shradhakhapra@Shradhas-Air ExpressDir %
```

```
shradhakhapra@Shradhas-Air ExpressDir % node index.js
<ref *1> [Function: app] {
  _events: [Object: null prototype] { mount: [Function:
onmount] },
  _eventsCount: 1,
  _maxListeners: undefined,
  setMaxListeners: [Function: setMaxListeners],
  getMaxListeners: [Function: getMaxListeners],
  emit: [Function: emit],
  addListener: [Function: addListener],
  on: [Function: addListener],
  prependListener: [Function: prependListener],
  once: [Function: once],
  prependOnceListener: [Function: prependOnceListener],
  removeListener: [Function: removeListener],
  off: [Function: removeListener],
  removeAllListeners: [Function: removeAllListeners],
  listeners: [Function: listeners],
  rawListeners: [Function: rawListeners],
  listenerCount: [Function: listenerCount],
  eventNames: [Function: eventNames],
  init: [Function: init],
  defaultConfiguration: [Function: defaultConfiguration]
},
  lazyrouter: [Function: lazyrouter],
  handle: [Function: handle],
  use: [Function: use],
  route: [Function: route],
  engine: [Function: engine],
  param: [Function: param],
  set: [Function: set],
  path: [Function: path],
  enabled: [Function: enabled],
  disabled: [Function: disabled],
  enable: [Function: enable],
  disable: [Function: disable],
  acl: [Function (anonymous)],
  bind: [Function (anonymous)],
  checkout: [Function (anonymous)],
  connect: [Function (anonymous)],
  copy: [Function (anonymous)]
}
```

```
JS index.js X
```

```
JS index.js > ...
```

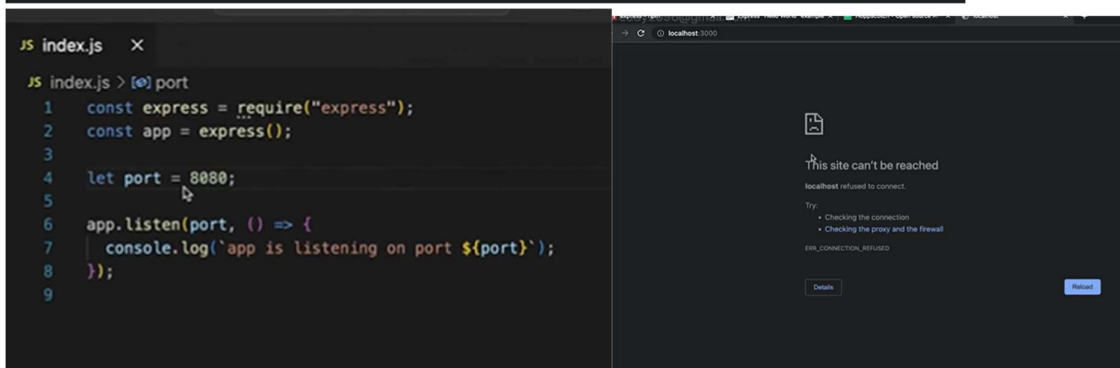
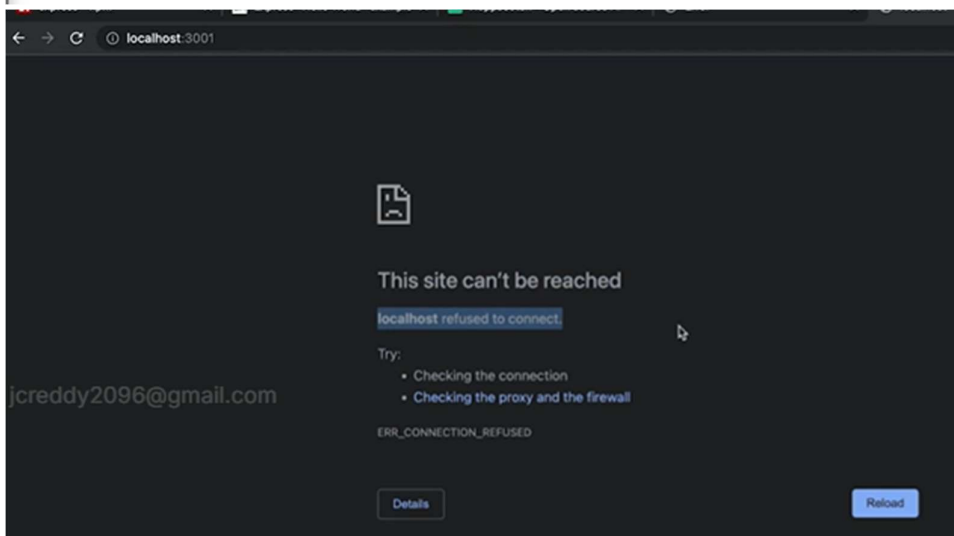
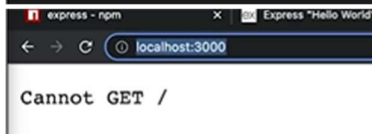
```
1 const express = require("express");
2 const app = express();
3
4 console.dir(app);
5
```

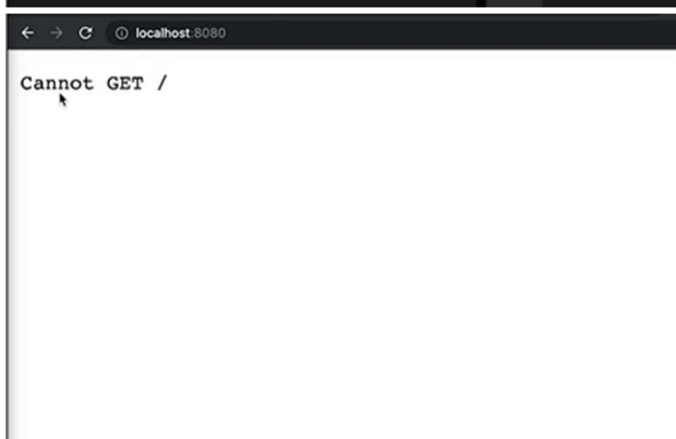
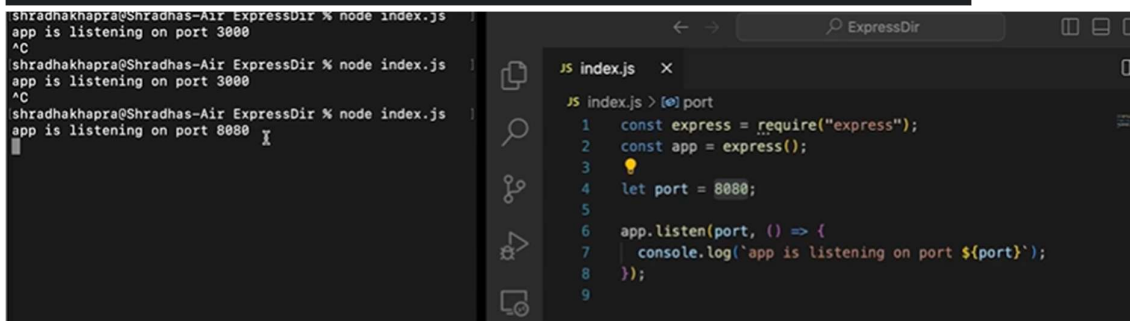
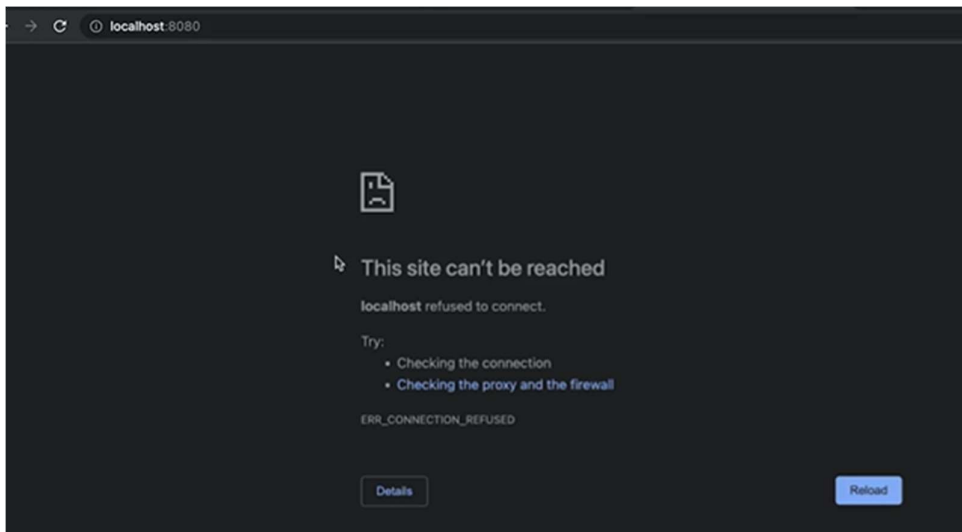
```
jcreddy2096@gmail.com
```

```
Spaces: 4 UTF-8 LF {} JavaScript Go Live
```

```
shradhakhapra@Shradhas-Air ExpressDir % node index.js
app is listening on port 3000
```

```
JS index.js x
JS index.js > app.listen() callback
1  const express = require("express");
2  const app = express();
3
4  let port = 3000;
5
6  app.listen(port, () => {
7    console.log(`app is listening on port ${port}`);
8  });
9
```





app.use

```
app.use((req, res) => {
  console.log("new incoming request");
});
```

express - npm | Express "Hello World" example | Hoppscotch - Open source API client | New Incognito tab

hoppscotch.io

APNA COLLEGE

HOPPSCOTCH

My Workspaces | Collections

REST | GraphQL | Realtime | Settings

Search | + New | Collections are empty | Add new

GET Untitled | http://localhost:8080/help | Send | Save

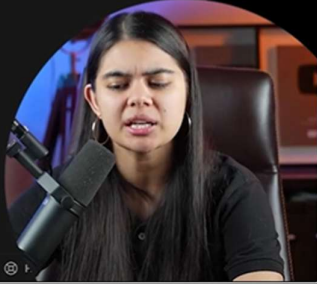
Parameters | Body | Headers | Authorization | Pre-request Script | Tests

Query Parameters

Parameter 1 | Value 1

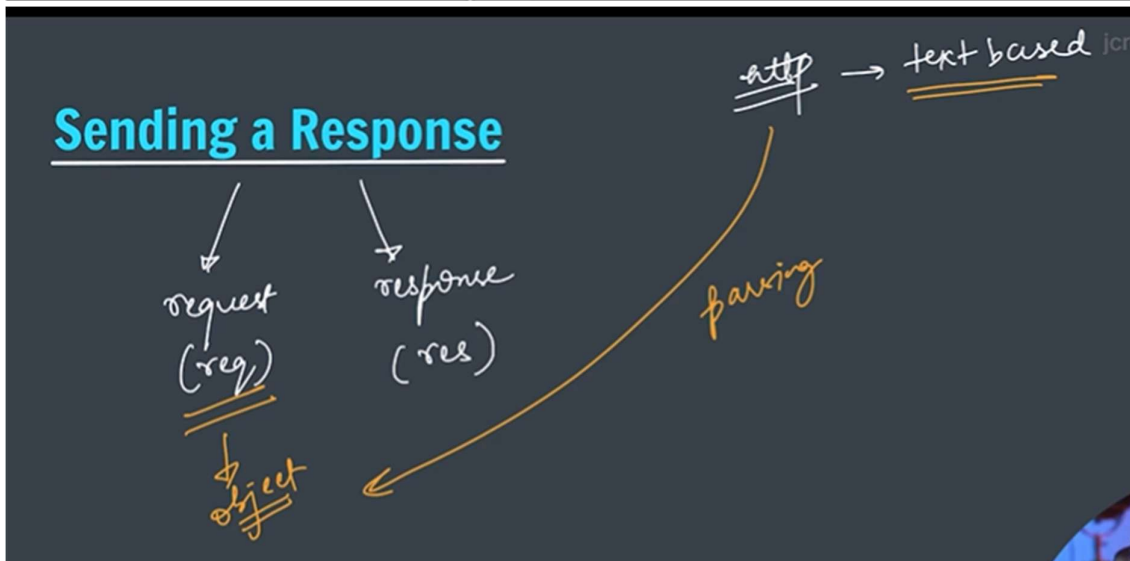
Send Request | Keyboard shortcuts | Search & command menu | Help menu | Documentation

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```
shradhakhapra@Shradhas-Air ExpressDir % node index.js
app is listening on port 8080
request received
request received
request received
```

```
JS index.js | JS index.js > ...
1 const express = require("express");
2 const app = express();
3
4 let port = 8080;
5
6 app.listen(port, () => {
7   console.log(`app is listening on port ${port}`);
8 });
9
10 app.use((req, res) => {
11   console.log("request received");
12 });
13
```

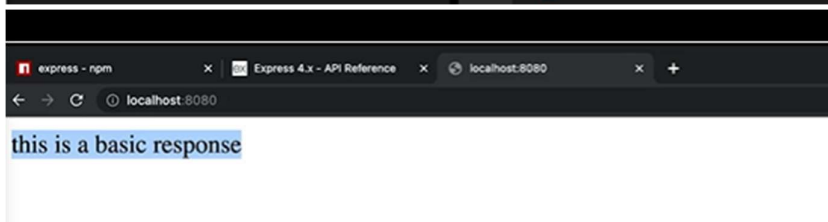


```
objectMode: false,
highWaterMark: 16384,
finalCalled: false,
needDrain: false,
ending: false,
ended: false,
finished: false,
destroyed: false,
decodeStrings: false,
defaultEncoding: 'utf8',
length: 0,
writing: false,
corked: 0,
sync: true,
bufferProcessing: false,
onwrite: [Function: bound onwrite],
writecb: null,
writelen: 0,
afterWriteTickInfo: null,
buffered: [],
bufferedIndex: 0,
allBuffers: true,
allNoop: true,
pendingcb: 0,
constructed: true,
prefinished: false,
errorEmitted: false,
emitClose: false,
autoDestroy: true,
errored: null,
closed: false,
closeEmitted: false,
[Symbol(kOnFinished)]: []
},
allowHalfOpen: true,
_socketname: null,
_pendingData: null,
_pendingEncoding: '',
server: Server {
  maxHeaderSize: undefined,
  insecureHTTPParser: undefined,
  _events: [Object: null prototype],
  _eventsCount: 2,
  _maxListeners: undefined,
  ...
}
```

```
JS index.js > app.use() callback
1  const express = require("express");
2  const app = express();
3
4  let port = 8080;
5
6  app.listen(port, () => {
7    console.log(`app is listening on port ${port}`);
8  });
9
10 app.use((req, res) => {
11   console.log(req);
12   console.log("request received");
13 });
14
```

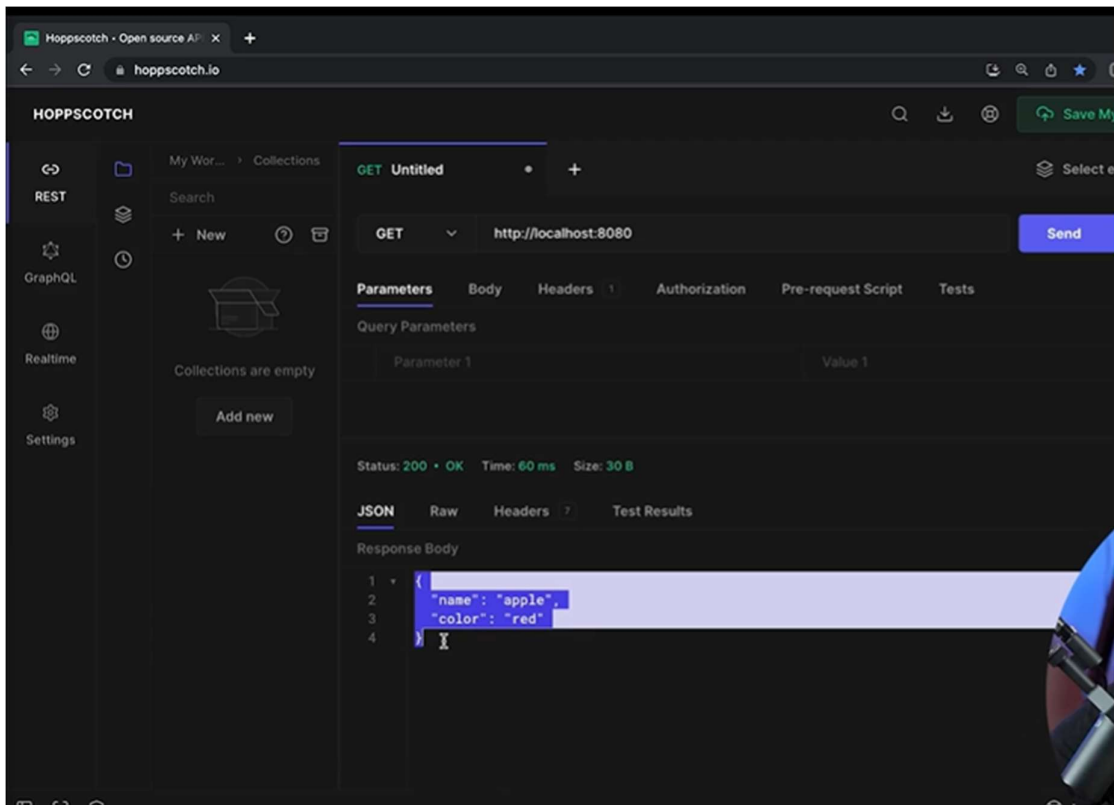
```
shradhakhapra@Shradhas-Air ExpressDir % node index.js
app is listening on port 8080
request received
request received
[]
```

```
JS index.js > app.use() callback
1  const express = require("express");
2  const app = express();
3
4  let port = 8080;
5
6  app.listen(port, () => {
7    console.log(`app is listening on port ${port}`);
8  });
9
10 app.use((req, res) => {
11   // console.log(req);
12   console.log("request received");
13   res.send("this is a basic response");
14 });
15
```



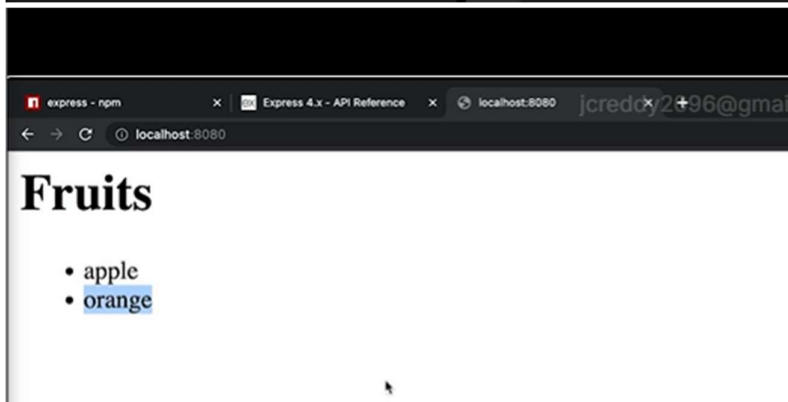
```
shradhakhapra@Shradhas-Air ExpressDir % node index.js
app is listening on port 8080
request received
request received
^C
shradhakhapra@Shradhas-Air ExpressDir % node index.js
app is listening on port 8080
```

```
JS index.js x
JS index.js > app.use() callback
1  const express = require("express");
2  const app = express();
3
4  let port = 8080;
5
6  app.listen(port, () => {
7    console.log(`app is listening on port ${port}`);
8  });
9
10 app.use((req, res) => {
11   console.log("request received");
12   res.send({
13     name: "apple",
14     color: "red",
15   });
16 });
17
```




```
shradhakhapra@Shradhas-Air ExpressDir % node index.js
app is listening on port 8080
request received
request received
^C
shradhakhapra@Shradhas-Air ExpressDir % node index.js
app is listening on port 8080
request received
request received
^C
shradhakhapra@Shradhas-Air ExpressDir % node index.js
app is listening on port 8080
request received
```

```
js index.js x
js index.js > app.use() callback
1  const express = require("express");
2  const app = express();
3
4  let port = 8080;
5
6  app.listen(port, () => {
7    console.log(`app is listening on port ${port}`);
8  });
9
10 app.use((req, res) => {
11   console.log("request received");
12   let code = "<h1>Fruits</h1> <ul><li>apple</li><li>orange";
13   res.send(code);
14 });
15
```



→ Routing

Routing

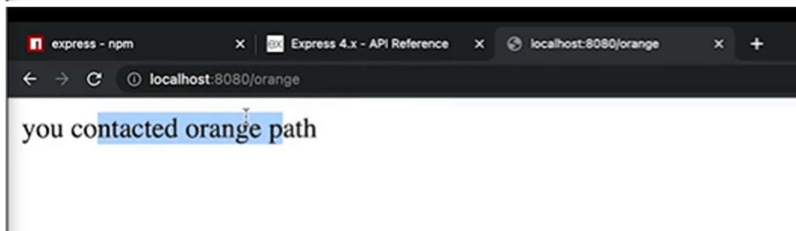
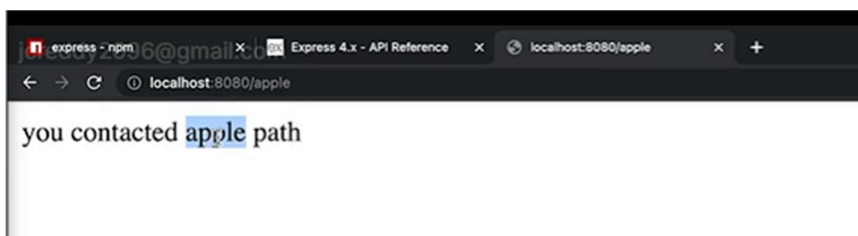
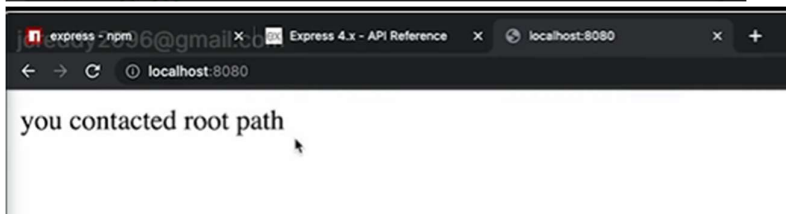
/search

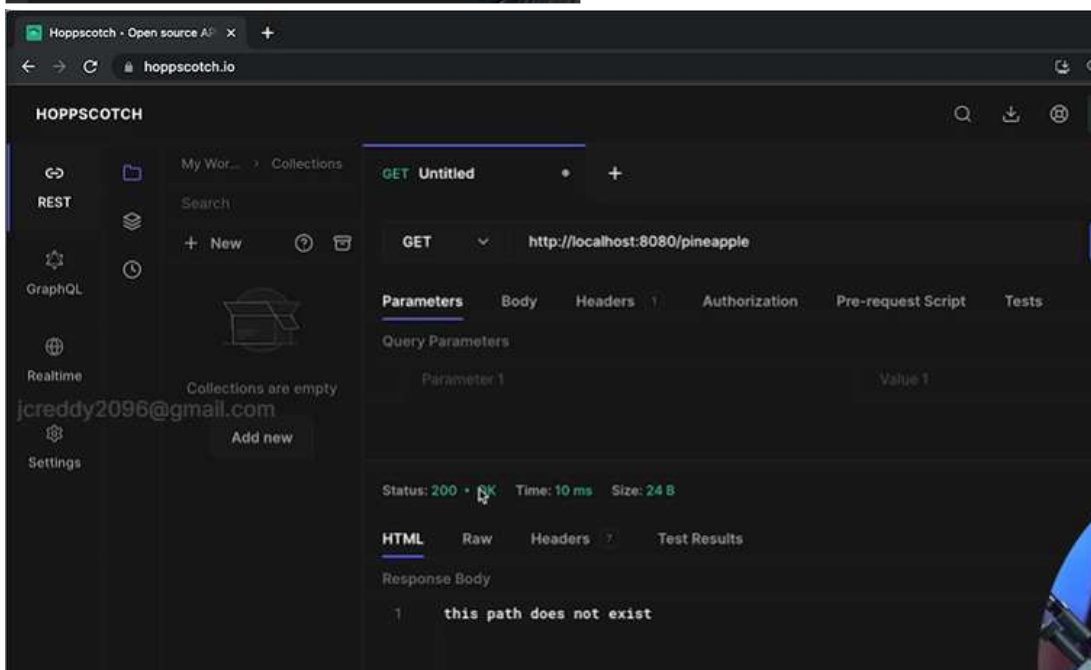
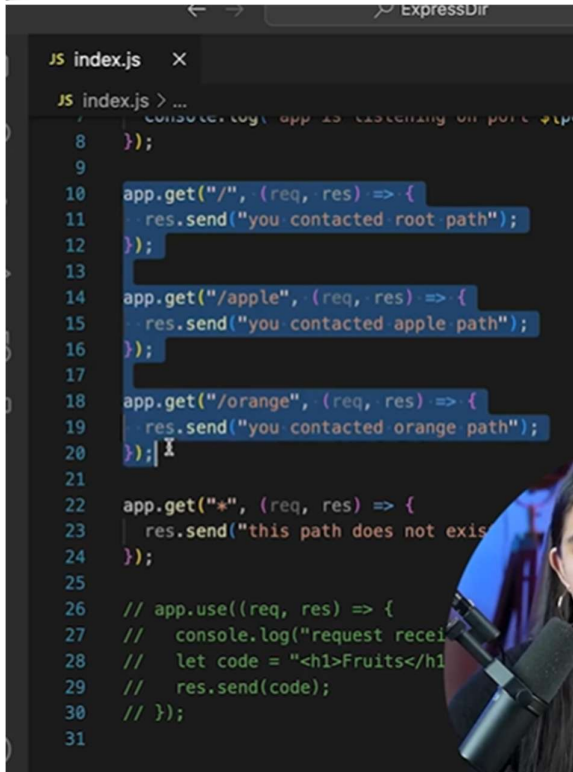
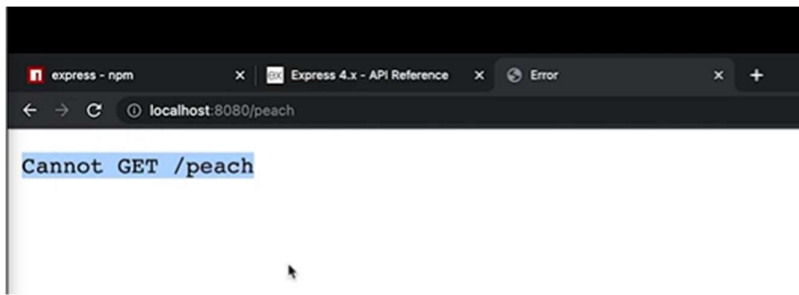
it is process of selecting a path for traffic in a network or between or across multiple networks.

```
app.get("/apple", (req, res) => {
  res.send({
    name: "apple",
    color: "red",
  });
});
```

jcreddy2096@gmail.com


```
JS index.js x
JS index.js > app.get("/orange") callback
1  const express = require("express");
2  const app = express();
3
4  let port = 8080;
5
6  app.listen(port, () => {
7    console.log(`app is listening on port ${port}`);
8  });
9
10 app.get("/", (req, res) => {
11   res.send("you contacted root path");
12 });
13
14 app.get("/apple", (req, res) => {
15   res.send("you contacted apple path");
16 });
17
18 app.get("/orange", (req, res) => {
19   res.send("you contacted orange path");
20 });
21
22 // app.use((req, res) => {
23 //   console.log("request received");
24 //   let code = "<h1>Fruits</h1> <ul><li>apple</li><li>ora
25 //   res.send(code);
26 // });
```





→Post

The top screenshot shows a code editor with the following JavaScript code in `index.js`:

```
13
14 app.get("/apple", (req, res) => {
15   res.send("you contacted apple path");
16 });
17
18 app.get("/orange", (req, res) => {
19   res.send("you contacted orange path");
20 });
21
22 app.get("*", (req, res) => {
23   res.send("this path does not exist");
24 });
25
26 app.post("/", (req, res) => {
27   res.send("you sent a post request to root");
28 });
29
30 // app.use((req, res) => {
```

The bottom screenshot shows the Hoppscotch REST client interface. The URL is `http://localhost:8080/` and the method is `POST`. The response status is `200 • OK`, with a time of `12 ms` and size of `31 B`. The response body is `you sent a post request to root`.

→Nodemon

Nodemon
To automatically restart server with code changes

→we need to install the nodemon globally

```
shradhakhapra@Shradhas-Air ExpressDir % npm install -g nodemon
added 34 packages, and audited 35 packages in 2s

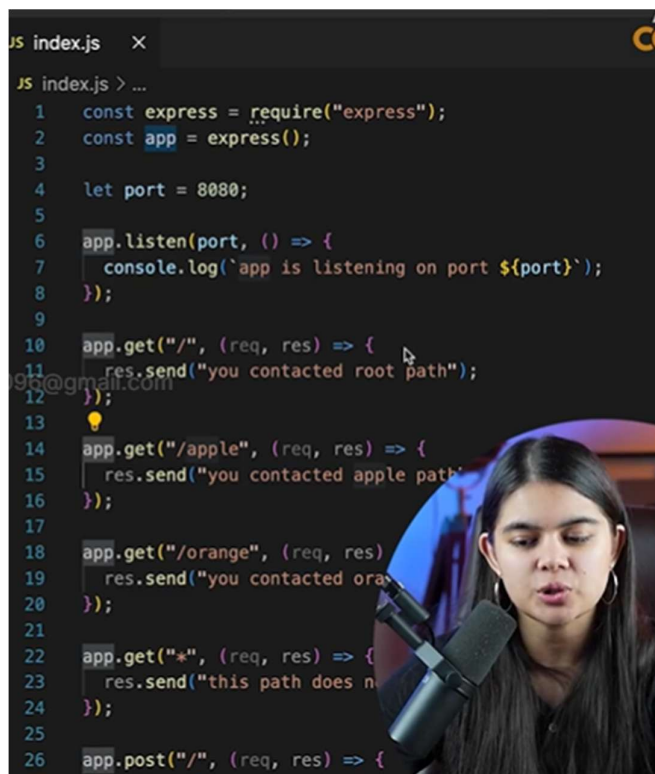
3 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
shradhakhapra@Shradhas-Air ExpressDir % nodemon -v
3.0.1
shradhakhapra@Shradhas-Air ExpressDir %
```

→to start the server

```
shradhakhapra@Shradhas-Air ExpressDir % nodemon index.js
[nodemon] 3.0.1
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node index.js`
app is listening on port 8080
```

→before



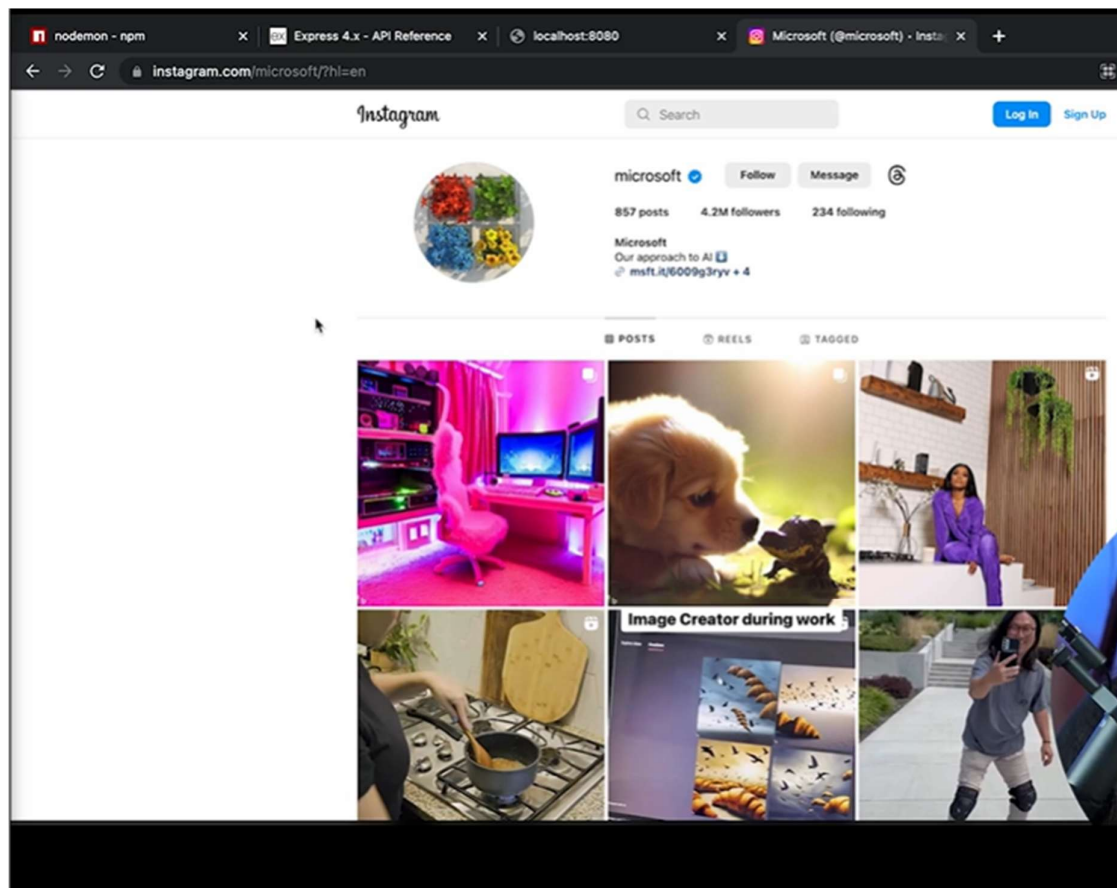
```
JS index.js x
JS index.js > ...
1  const express = require("express");
2  const app = express();
3
4  let port = 8080;
5
6  app.listen(port, () => {
7    console.log(`app is listening on port ${port}`);
8  });
9
10 app.get("/", (req, res) => {
11   res.send("you contacted root path");
12 });
13
14 app.get("/apple", (req, res) => {
15   res.send("you contacted apple path");
16 });
17
18 app.get("/orange", (req, res) => {
19   res.send("you contacted orange path");
20 });
21
22 app.get("*", (req, res) => {
23   res.send("this path does not exist");
24 });
25
26 app.post("/", (req, res) => {
```

→after

```
shradhakhapra@Shradhas-Air ExpressDir % npm install -g nodemon
added 34 packages, and audited 35 packages in 2s
3 packages are looking for funding
  run 'npm fund' for details
found 0 vulnerabilities
shradhakhapra@Shradhas-Air ExpressDir % nodemon -v
3.0.1
shradhakhapra@Shradhas-Air ExpressDir % nodemon index.js
[nodemon] 3.0.1
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting 'node index.js'
app is listening on port 8080
[nodemon] restarting due to changes...
[nodemon] starting 'node index.js'
app is listening on port 8080

JS index.js
1 const express = require("express");
2 const app = express();
3
4 let port = 8080;
5
6 app.listen(port, () => {
7   console.log(`app is listening on port ${port}`);
8 });
9
10 app.get("/", (req, res) => {
11   res.send("hello, i am root");
12 });
13
14 app.get("/apple", (req, res) => {
15   res.send("you contacted apple path");
16 });
17
18 app.get("/orange", (req, res) => {
19   res.send("you contacted orange path");
20 });
21
22 app.get("/*", (req, res) => {
23   res.send("this path does not exist");
24 });
25
26 app.post("/", (req, res) => {
```

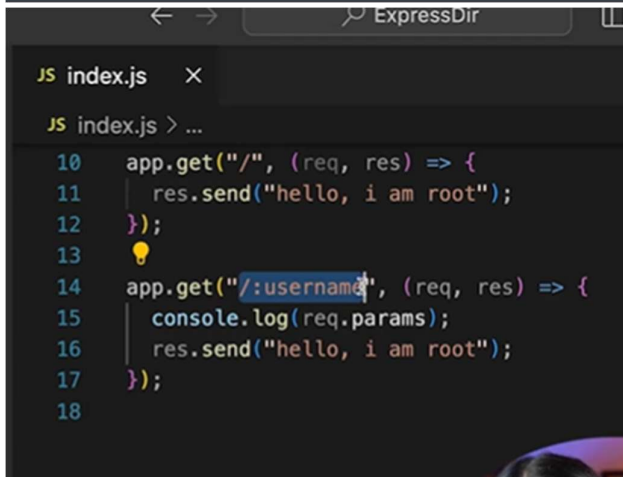
→ Path parameters



Path Parameters

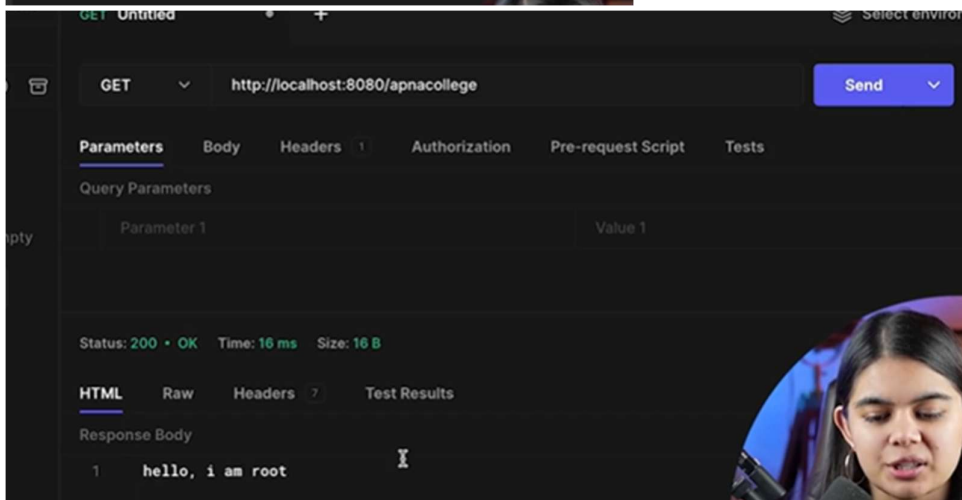
req.params

```
app.get("/ig/:username", (req, res) => {  
  let { username } = req.params;  
  res.send(`This account belongs to @${username}`);  
});
```



A screenshot of a code editor window titled 'index.js'. The code defines two routes. The first route is for the root path ('/'). The second route is for '/:username', where the parameter ':username' is highlighted with a blue selection box. A lightbulb icon is visible above the second route definition.

```
10 app.get("/", (req, res) => {  
11   res.send("hello, i am root");  
12 });  
13   
14 app.get("/:username", (req, res) => {  
15   console.log(req.params);  
16   res.send("hello, i am root");  
17 });  
18
```



A screenshot of a REST client interface. The top bar shows a GET request to 'http://localhost:8080/apnacollege'. Below the bar, the 'Parameters' tab is active, showing an empty table with columns 'Parameter 1' and 'Value 1'. The status bar indicates 'Status: 200 • OK', 'Time: 16 ms', and 'Size: 16 B'. The 'HTML' tab is selected, showing the response body 'hello, i am root'.

GET http://localhost:8080/apnacollege

Parameters

Parameter 1	Value 1
-------------	---------

Status: 200 • OK Time: 16 ms Size: 16 B

HTML

Response Body

```
1 hello, i am root
```



```
shradhakhapra@Shradhas-Air ExpressDir % npm install -g nodemon
added 34 packages, and audited 35 packages in 2s

3 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
shradhakhapra@Shradhas-Air ExpressDir % nodemon -v
3.0.1
shradhakhapra@Shradhas-Air ExpressDir % nodemon index.js

[nodemon] 3.0.1
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting 'node index.js'
app is listening on port 8080
[nodemon] restarting due to changes...
[nodemon] starting 'node index.js'
app is listening on port 8080
[nodemon] restarting due to changes...
[nodemon] starting 'node index.js'
app is listening on port 8080
[nodemon] restarting due to changes...
[nodemon] starting 'node index.js'
app is listening on port 8080
{ username: 'apnacollege' }
```

JS index.js

```
JS index.js > app.get("/:username") callback
10 app.get("/", (req, res) => {
11   res.send("hello, i am root");
12 });
13
14 app.get("/:username", (req, res) => {
15   console.log(req.params);
16   res.send("hello, i am root");
17 });
18
```

```
shradhakhapra@Shradhas-Air ExpressDir % npm install -g nodemon
added 34 packages, and audited 35 packages in 2s

3 packages are looking for funding
  run `npm fund` for details

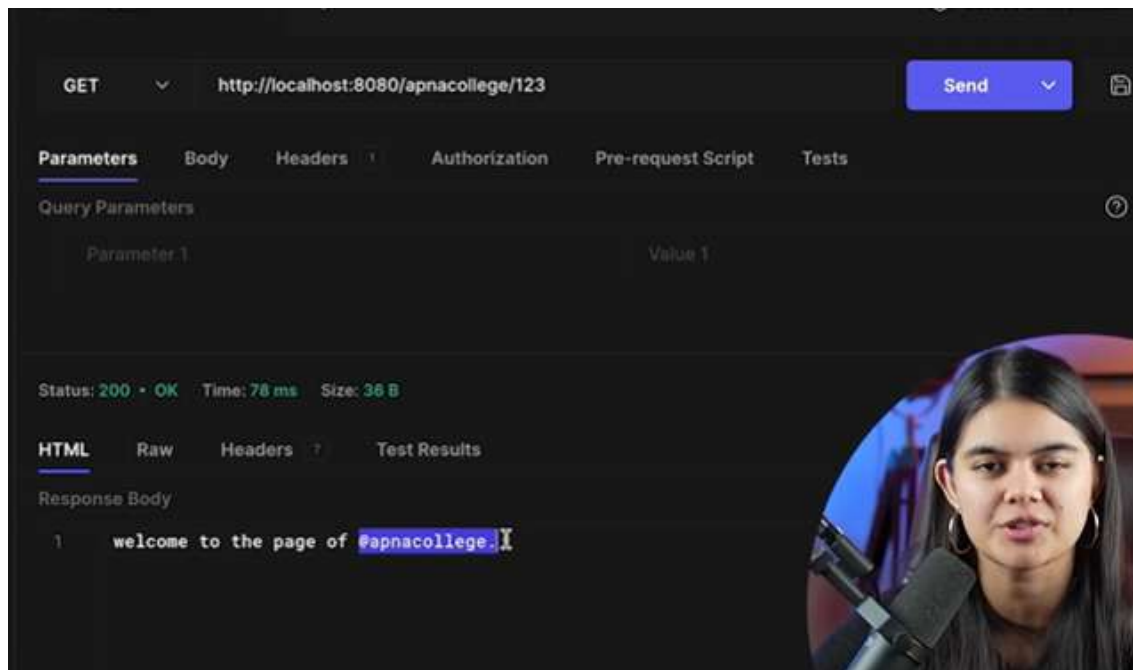
found 0 vulnerabilities
shradhakhapra@Shradhas-Air ExpressDir % nodemon -v
3.0.1
shradhakhapra@Shradhas-Air ExpressDir % nodemon index.js

[nodemon] 3.0.1
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting 'node index.js'
app is listening on port 8080
[nodemon] restarting due to changes...
[nodemon] starting 'node index.js'
app is listening on port 8080
[nodemon] restarting due to changes...
[nodemon] starting 'node index.js'
app is listening on port 8080
[nodemon] restarting due to changes...
[nodemon] starting 'node index.js'
app is listening on port 8080
{ username: 'apnacollege' }
[nodemon] restarting due to changes...
[nodemon] starting 'node index.js'
app is listening on port 8080
{ username: 'apnacollege', id: '123' }
```

JS index.js

```
JS index.js > ...
10 app.get("/", (req, res) => {
11   res.send("hello, i am root");
12 });
13
14 app.get("/:username/:id", (req, res) => {
15   console.log(req.params);
16   res.send("hello, i am root");
17 });
18
```

```
index.js > app.get("/:username/:id") callback
0 app.get("/", (req, res) => {
1   res.send("hello, i am root");
2 });
3
4 app.get("/:username/:id", (req, res) => {
5   let { username, id } = req.params;
6   res.send(`welcome to the page of ${username}.`);
7 });
8
```

```
JS index.js x
JS index.js > app.get("/:username/:id") callback
10 app.get("/", (req, res) => {
11   res.send("hello, i am root");
12 });
13
14 app.get("/:username/:id", (req, res) => {
15   let { username, id } = req.params;
16   let htmlStr = `<h1>welcome to the page of @${username}!<
17   res.send(htmlStr);
18 });
19
```



→Query Strings

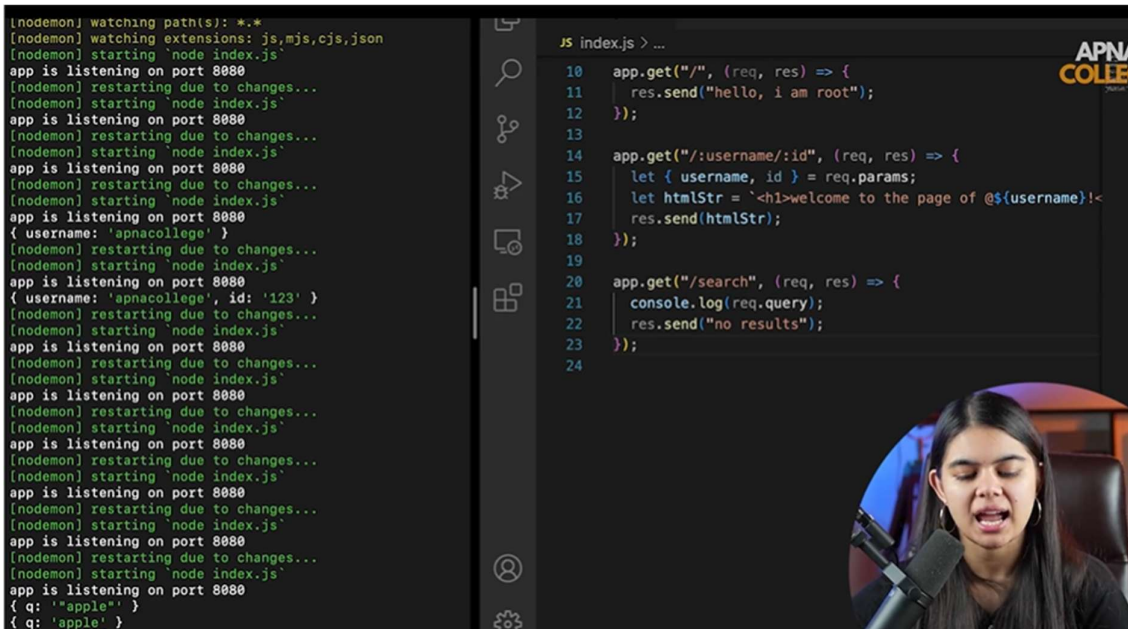
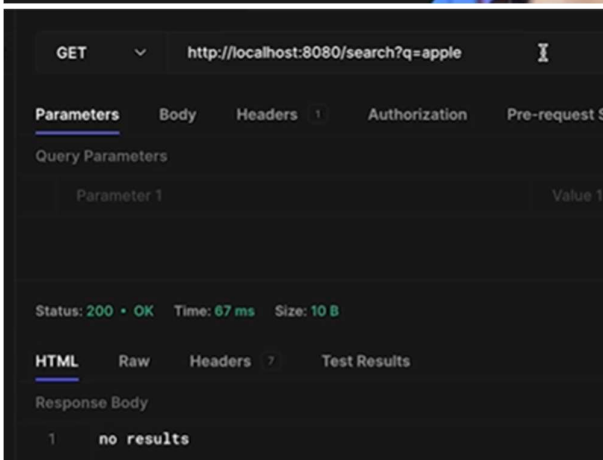
```

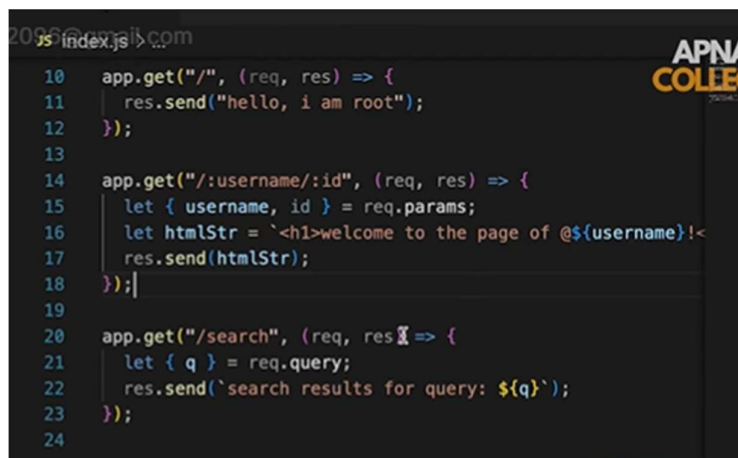
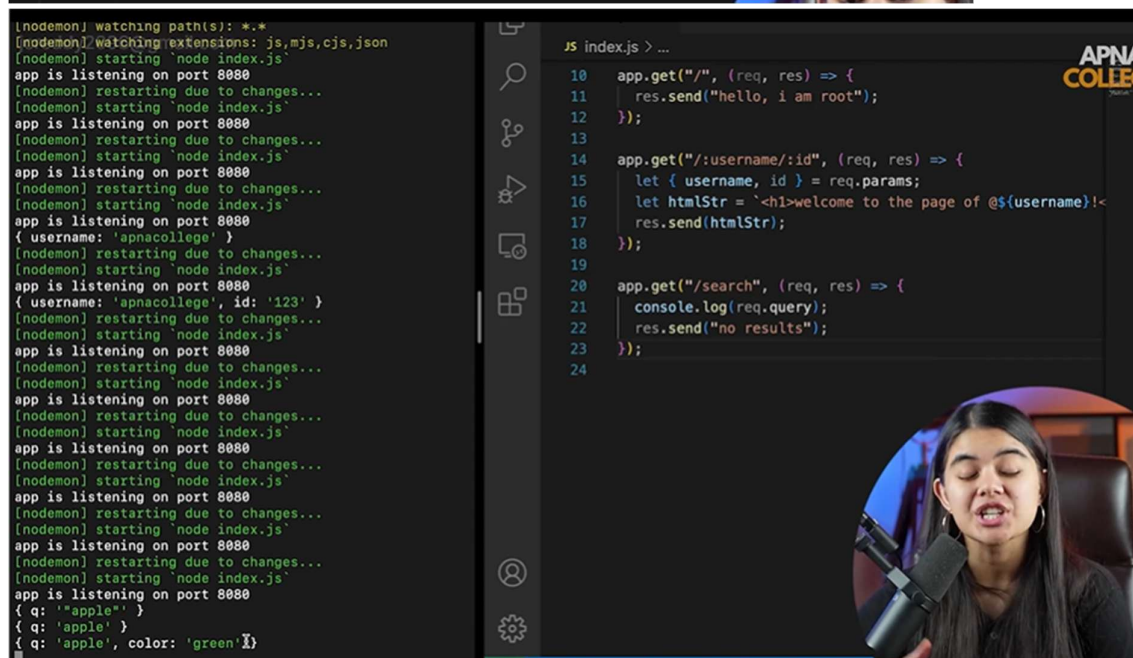
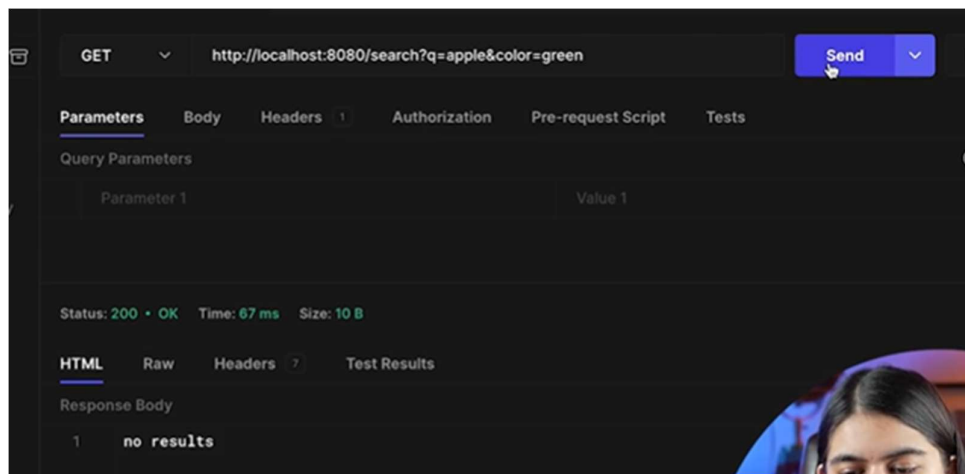
app.get("/", (req, res) => {
  res.send("hello, i am root");
});

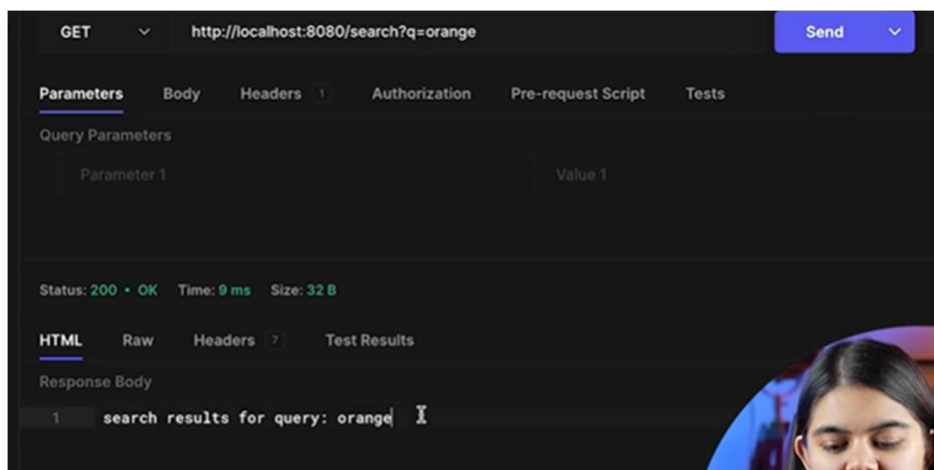
app.get("/:username/:id", (req, res) => {
  let { username, id } = req.params;
  let htmlStr = `<h1>welcome to the page of @${username}!<
  res.send(htmlStr);
});

app.get("/search", (req, res) => {
  console.log(req.query);
  res.send("no results");
});

```







```
app.get("/search", (req, res) => {  
  let { q } = req.query;  
  if(!q) {  
    // ...  
  }  
  res.send(`<h1>search results for query: ${q}</h1>`);  
});
```

```
app.get("/search", (req, res) => {  
  send(body?: any): Response<any, Record<string, any>, number>  
  res.send("<h1>nothing searched</h1>");  
  res.send(`<h1>search results for query: ${q}</h1>`);  
});
```

